



9535 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
HD 32297 - NIRSpec IFU				
	1	HD 32297 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(1) HD-32297
	2	HD 32297 PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(3) HD-30365-PSF-CALIBRATOR
HD 15745 - NIRSpec IFU				

JWST Proposal 9535 (Created: Friday, March 13, 2026, 12:00:51PM Eastern Standard Time) - Overview

Folder	Observation	Label	Observing Template	Science Target
	3	HD 15745 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(5) HD-15745
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HD 15115 - NIRSpec IFU				
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	6	HD 15115 PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(11) HD-16647-PSF-CALIBRATOR
HD 141943 - NIRSpec IFU				
	7	HD 141943 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(13) HD-141943
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HD 61005 - NIRSpec IFU - roll1				
	15	HD 61005 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(21) HD-61005
	16	HD 61005 PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(23) HD-56161-PSF-CALIBRATOR
HD 61005 - NIRSpec IFU - roll2				
	17	HD 61005 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(21) HD-61005
	18	HD 61005 PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(23) HD-56161-PSF-CALIBRATOR
AU Mic - NIRSpec IFU				
	11	AU Mic Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(25) AU-MIC
	12	AU Mic PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(27) HD-192855-PSF-CALIBRATOR
HD 202917 - NIRSpec IFU				
	13	HD 202917 Debris Disk - NIRSpec IFU	NIRSpec IFU Spectroscopy	(17) HD-202917
	14	HD 202917 PSF Calibrator - NIRSpec IFU	NIRSpec IFU Spectroscopy	(19) HD-199201-PSF-CALIBRATOR

ABSTRACT

We propose to obtain near-infrared reflectance spectra from dust in the Kuiper Belt regions of seven nearby debris disks. During the past year, JWST has provided exquisite scattered light spectroscopy that has revealed the presence of water ice in two debris disk systems in the beta Pic Moving Group: HD 181327 and beta Pic itself. Despite the similarity in age, the water ice properties are surprisingly different with evidence for large, crystalline grains around HD 181327 and amorphous water ice around beta Pic. One explanation for the presence of crystalline water ice in HD 181327 is high velocity collisions among the planetesimals induced by planet migration. We propose to increase the sample of debris disks studied using NIRSpec IFU spectroscopy from three to ten objects to verify that water ice is common and understand its lifecycle in extrasolar planetary systems.

OBSERVING DESCRIPTION

We propose to obtain NIRSpec IFU observations of seven debris disks that appear spatially extended in scattered light that have already measured near-infrared surface brightnesses from HST/NICMOS. High SNR, low resolution ($R \sim 100$), NIRSpec IFU observations at 0.6 - 5.3 micron will enable (1) sensitive searches for solid-state reflectance features at 3.0 and 4.2 micron from volatile frosts such as H₂O and CO₂, (2) detailed constraints on the grain composition and size from the wavelengths and shapes of the solid-state features and the color of the scattered light, and (3) mapping of the particle composition and size as a function of position in the disk.

To mitigate the effect of the bright host star on the detection of the disk, we plan to point the IFU so that the host star is an average of 0.5" - 0.75" off the IFU FOV and to apply reference PSF subtraction techniques to the data. Ground-based testing of the NIRSpec instrument indicates that the IFU will not suffer from scattered light from objects just outside the FOV. We have selected reference stars that are located near our targets (within 20 deg on the sky) and have similar spectral types. To facilitate PSF subtraction, we request Wide Aperture Target Acquisition and a 9-point Small Cycling dither pattern. Dithering will enable reconstruction of the PSF from spatially subsampled data. Observations of a target and its corresponding PSF reference should be observed consecutively using a non-interrupt.

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Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HD-32297	RA: 05 02 27.4418 (75.6143408d) Dec: +07 27 39.31 (7.46092d) Equinox: J2000	Proper Motion RA: 3.874805427246334E-4 sec of time/yr Proper Motion Dec: -0.02377200009959779 arcsec/yr Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Circumstellar disks, Circumstellar dust, Circumstellar gas, Debris disks] Extended=YES				
	(2)	OFFSET-TA-STAR-FOR-HD-32297	RA: 05 02 26.2506 (75.6093775d) Dec: +07 27 54.70 (7.46519d) Equinox: J2000	Proper Motion RA: 0.841313519 mas/yr Proper Motion Dec: 0.272100914 mas/yr Epoch of Position: 2016	
	Comments: This is the only offset star for use in WATA TA for HD 32297. Coordinates from DR3, Gaia DR3 3289902892417327104. Gaia RUWE = 1.03, consistent with a good measurement of a single star. Gaia magnitudes G = 17.58, Bp = 17.93, Rp = 17.03 Category=Star Description=[G stars] Extended=NO				
	(3)	HD-30365-PSF-CALIBRATOR	RA: 04 47 17.9234 (71.8246808d) Dec: +05 47 17.59 (5.78822d) Equinox: J2000	Proper Motion RA: 14.674000000000001 mas/yr Proper Motion Dec: -17.857999955595005 mas/yr Parallax: 0.005368" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[A stars, Point spread function]				
	(4)	OFFSET-TA-STAR-FOR-HD-30365	RA: 04 47 15.6225 (71.8150938d) Dec: +05 47 33.00 (5.79250d) Equinox: J2000	Proper Motion RA: 0.952326404 mas/yr Proper Motion Dec: -2.705509483 mas/yr Epoch of Position: 2016	
	Comments: This is the closet star to use in WATA TA for HD-30365. Coordinates from DR3, Gaia DR3 3281991841830087680. Gaia RUWE = 1.01, consistent with a good measurement of a single star. Gaia magnitudes G = 19.61, Bp = 20.27, Rp = 18.90 Category=Star Description=[K stars] Extended=NO				
	(5)	HD-15745	RA: 02 32 55.8109 (38.2325454d) Dec: +37 20 1.04 (37.33362d) Equinox: J2000	Proper Motion RA: 46.436 mas/yr Proper Motion Dec: -47.90300001786818 mas/yr Parallax: 0.0139411" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, F stars]					

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(6)	OFFSET-TA-STAR-FOR-HD- 15745 RA: 02 32 55.1012 (38.2295883d) Dec: +37 19 49.31 (37.33036d) Equinox: J2000	Proper Motion RA: 6.939120407 mas/yr Proper Motion Dec: 6.907474425 mas/yr Epoch of Position: 2016
Comments: This is the closet star to use in WATA TA for HD-15745		
Coordinates from DR3, Gaia DR3 333649483509999744.		
Gaia RUWE = 1.12, consistent with a good measurement of a single star.		
Gaia magnitudes G = 18.24, Bp = 18.8, Rp = 17.16		
Category=Star		
Description=[M stars]		
Extended=NO		
(7)	HD-19301-PSF-CALIBRATOR RA: 03 07 32.0857 (46.8836904d) Dec: +39 05 12.14 (39.08671d) Equinox: J2000	Proper Motion RA: -27.145 mas/yr Proper Motion Dec: -28.295000083744526 mas/yr Parallax: 0.0100366" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.		
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.		
Category=Calibration		
Description=[Point spread function]		
(8)	OFFSET-TA-STAR-FOR-HD- 19301 RA: 03 07 30.7283 (46.8780346d) Dec: +39 05 28.25 (39.09118d) Equinox: J2000	Proper Motion RA: -0.484 mas/yr Proper Motion Dec: -1.955 mas/yr Epoch of Position: 2016
Comments: This is the closet star to use in WATA TA for HD-19301		
Coordinates from DR3, Gaia DR3 142695654140064384.		
Gaia RUWE = 1.152, consistent with a good measurement of a single star.		
Gaia magnitudes G = 18.02, Bp = 18.55, Rp = 18.51		
Category=Star		
Description=[A stars]		
Extended=NO		
(9)	HD-15115 RA: 02 26 16.2458 (36.5676908d) Dec: +06 17 33.19 (6.29255d) Equinox: J2000	Proper Motion RA: 88.013 mas/yr Proper Motion Dec: -50.41899994466803 mas/yr Parallax: 0.020502600000000003" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.		
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.		
Category=Star		
Description=[Circumstellar disks, Circumstellar dust, Debris disks]		

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(10)	OFFSET-TA-STAR-FOR-HD-15115	RA: 02 26 15.1371 (36.5630713d) Dec: +06 17 53.33 (6.29815d) Equinox: J2000	Proper Motion RA: 4.827436376 mas/yr Proper Motion Dec: -9.483733495 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for HD-15115</i> <i>Coordinates from DR3, Gaia DR3 2517397945572650240.</i> <i>Gaia RUWE = 1.06, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 20.64, Bp = 20.94, Rp = 19.52</i> <i>Category=Star</i> <i>Description=[K stars]</i> <i>Extended=NO</i>			
(11)	HD-16647-PSF-CALIBRATOR	RA: 02 40 15.6918 (40.0653825d) Dec: +06 06 43.02 (6.11195d) Equinox: J2000	Proper Motion RA: 49.263 mas/yr Proper Motion Dec: -1.2180000112493872 mas/yr Parallax: 0.0254832" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[Point spread function]</i>			
(12)	OFFSET-TA-STAR-FOR-HD-16647	RA: 02 40 13.7482 (40.0572842d) Dec: +06 06 43.44 (6.11207d) Equinox: J2000	Proper Motion RA: 42.02982583 mas/yr Proper Motion Dec: -48.15819497 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for HD-16647.</i> <i>Coordinates from DR3, Gaia DR3 18265778650907776.</i> <i>Gaia RUWE = 1.05, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 19.00, Bp = 17.87, Rp = 17.87</i> <i>Category=Star</i> <i>Description=[M stars]</i> <i>Extended=NO</i>			
(13)	HD-141943	RA: 15 53 27.2917 (238.3637154d) Dec: -42 16 0.71 (-42.26686d) Equinox: J2000	Proper Motion RA: -43.236 mas/yr Proper Motion Dec: -65.58800002949283 mas/yr Parallax: 0.0166271" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Circumstellar disks, Circumstellar dust, Debris disks, G stars]</i>			

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(14)	OFFSET-TA-STAR-FOR-HD-141943	RA: 15 53 26.8595 (238.3619146d) Dec: -42 15 51.37 (-42.26427d) Equinox: J2000	Proper Motion RA: -8.167734119 mas/yr Proper Motion Dec: -9.614247932 mas/yr Epoch of Position: 2016
<i>Comments: This is one possible offset star for use in WATA TA for HD-141943. A few other candidates are available and could be used if it eases scheduling.</i> <i>Coordinates from DR3, Gaia DR3 5995577880528197632.</i> <i>Gaia RUWE = 1.01, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 18.51, Bp = 18.79, Rp = 17.56</i> <i>Category=Star</i> <i>Description=[K stars]</i> <i>Extended=NO</i>			
(15)	HD-141885-PSF-CALIBRATOR	RA: 15 52 52.0812 (238.2170050d) Dec: -37 23 54.91 (-37.39859d) Equinox: J2000	Proper Motion RA: 3.7680000000000002 mas/yr Proper Motion Dec: -30.806999984633876 mas/yr Parallax: 0.0096624" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[G stars, Point spread function]</i>			
(16)	OFFSET-TA-STAR-FOR-HD-141885	RA: 15 52 53.3339 (238.2222246d) Dec: -37 23 56.06 (-37.39891d) Equinox: J2000	Proper Motion RA: -5.570341212 mas/yr Proper Motion Dec: -5.417355646 mas/yr Epoch of Position: 2016
<i>Comments: This is one possible offset star for use in WATA TA for HD-141885. A few other candidates are available and could be used if it eases scheduling.</i> <i>Coordinates from DR3, Gaia DR3 6010361192321029504.</i> <i>Gaia RUWE = 0.97, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 15.67, Bp = 16.3, Rp = 14.91</i> <i>Category=Star</i> <i>Description=[K stars]</i> <i>Extended=NO</i>			
(17)	HD-202917	RA: 21 20 49.9576 (320.2081567d) Dec: -53 02 3.16 (-53.03421d) Equinox: J2000	Proper Motion RA: 31.426999999999996 mas/yr Proper Motion Dec: -95.58700010074972 mas/yr Parallax: 0.021409300000000003" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Circumstellar disks, Circumstellar dust, Debris disks, G dwarfs]</i>			

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(18)	OFFSET-TA-STAR-FOR-HD- 202917	RA: 21 20 51.2625 (320.2135937d) Dec: -53 01 42.40 (-53.02844d) Equinox: J2000	Proper Motion RA: -1.749635345 mas/yr Proper Motion Dec: -12.79898866 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for HD-202917</i> <i>Coordinates from DR3, Gaia DR3 6463782431652157056.</i> <i>Gaia RUWE = 0.98, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 16.69, Bp = 16.98 , Rp = 16.12</i> <i>Category=Star</i> <i>Description=[G stars]</i> <i>Extended=NO</i>			
(19)	HD-199201-PSF- CALIBRATOR	RA: 20 58 11.9458 (314.5497742d) Dec: -58 30 18.58 (-58.50516d) Equinox: J2000	Proper Motion RA: -40.535 mas/yr Proper Motion Dec: -81.06800003133685 mas/yr Parallax: 0.0117922" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[G stars, Point spread function]</i>			
(20)	OFFSET-TA-STAR-FOR-HD- 199201	RA: 20 58 13.1022 (314.5545925d) Dec: -58 30 32.94 (-58.50915d) Equinox: J2000	Proper Motion RA: -3.719020341 mas/yr Proper Motion Dec: -15.26260054 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for HD-199201</i> <i>Coordinates from DR3, Gaia DR3 6456355819017372800</i> <i>Gaia RUWE = 0.96, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 18.48, Bp = 18.71 , Rp =17.97</i> <i>Category=Star</i> <i>Description=[G stars]</i> <i>Extended=NO</i>			
(21)	HD-61005	RA: 07 35 47.4624 (113.9477600d) Dec: -32 12 14.05 (-32.20390d) Equinox: J2000	Proper Motion RA: -55.05 mas/yr Proper Motion Dec: 74.319 mas/yr Parallax: 0.0274344" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Star</i> <i>Description=[Circumstellar disks, Circumstellar dust, Debris disks, G dwarfs]</i>			

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(22)	OFFSET-TA-STAR-FOR-HD-61005	RA: 07 35 47.6035 (113.9483479d) Dec: -32 12 2.58 (-32.20072d) Equinox: J2000	Proper Motion RA: -2.150659522 mas/yr Proper Motion Dec: 2.938559635 mas/yr Epoch of Position: 2016
<i>Comments: This is one possible offset star for use in WATA TA for HD-61005. A few other candidates are available and could be used if it eases scheduling.</i> <i>Coordinates from DR3, Gaia DR3 5592237707928068096.</i> <i>Gaia RUWE = 1.13, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 16.27, Bp = 16.66 , Rp = 15.54</i> Category=Star Description=[G stars]			
(23)	HD-56161-PSF-CALIBRATOR	RA: 07 14 36.7020 (108.6529250d) Dec: -30 39 17.63 (-30.65490d) Equinox: J2000	Proper Motion RA: -83.772 mas/yr Proper Motion Dec: 45.286 mas/yr Parallax: 0.0082534" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Calibration Description=[Point spread function]			
(24)	OFFSET-TA-STAR-FOR-HD-56161	RA: 07 14 37.5648 (108.6565200d) Dec: -30 38 53.59 (-30.64822d) Equinox: J2000	Proper Motion RA: 3.479251852 mas/yr Proper Motion Dec: -5.762613576 arcsec/yr Epoch of Position: 2016
<i>Comments: This is one possible offset star for use in WATA TA for HD-56161. A few other candidates are available and could be used if it eases scheduling.</i> <i>Coordinates from DR3, Gaia DR3 5605917384922973440.</i> <i>Gaia RUWE = 0.97, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 14.04, Bp = 14.51 , Rp = 13.42</i> Category=Star Description=[K stars] Extended=NO			
(25)	AU-MIC	RA: 20 45 9.5325 (311.2897188d) Dec: -31 20 27.24 (-31.34090d) Equinox: J2000	Proper Motion RA: 281.319 mas/yr Proper Motion Dec: -360.1480000497759 mas/yr Parallax: 0.1029432" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, Exoplanet Systems, M dwarfs]			

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(26)	OFFSET-TA-STAR-FOR-AU-MIC	RA: 20 45 11.5470 (311.2981125d) Dec: -31 20 30.65 (-31.34185d) Equinox: J2000	Proper Motion RA: 6.739083093 mas/yr Proper Motion Dec: -7.18877304 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for AU-MIC</i> <i>Coordinates from DR3, Gaia DR3 6794047588308518144</i> <i>Gaia RUWE = 1.04, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 16.59, Bp = 17.07, Rp =15.93</i> <i>Category=Star</i> <i>Description=[K stars]</i> <i>Extended=NO</i>			
(27)	HD-192855-PSF-CALIBRATOR	RA: 20 18 6.0466 (304.5251942d) Dec: -26 13 56.33 (-26.23231d) Equinox: J2000	Proper Motion RA: 8.631 mas/yr Proper Motion Dec: 3.01 mas/yr Parallax: 0.001062" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Calibration</i> <i>Description=[Point spread function]</i>			
(28)	OFFSET-TA-STAR-FOR-HD-192855	RA: 20 18 7.5054 (304.5312725d) Dec: -26 13 53.56 (-26.23154d) Equinox: J2000	Proper Motion RA: 4.049 mas/yr Proper Motion Dec: -10.243 mas/yr Epoch of Position: 2016
<i>Comments: This is the closet star to use in WATA TA for HD 192855</i> <i>Coordinates from DR3, Gaia DR3 6847316177891818880</i> <i>Gaia RUWE = 0.975, consistent with a good measurement of a single star.</i> <i>Gaia magnitudes G = 18.78, Bp = 19.29, Rp =18.12</i> <i>Category=Star</i> <i>Description=[G stars]</i> <i>Extended=NO</i>			

Proposal 9535 - Observation 1 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 1: HD 32297 Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	HD-32297	RA: 05 02 27.4418 (75.6143408d) Dec: +07 27 39.31 (7.46092d) Equinox: J2000			Proper Motion RA: 3.874805427246334E-4 sec of time/yr Proper Motion Dec: -0.02377200009959779 arcsec/yr Epoch of Position: 2015.5						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Circumstellar disks, Circumstellar dust, Circumstellar gas, Debris disks] Extended=YES											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 OFFSET-TA-STAR-FOR-HD-32297	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	5	1	25.0		0.0		-20.0		0.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Proposal 9535 - Observation 1 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 56 to 77 Degrees (V3 277.02835083 to 298.02835083) Sequence Observations 1, 2, Non-interruptible
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Proposal 9535 - Observation 2 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 2: HD 32297 PSF Calibrator - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(HD 32297 PSF Calibrator - NIRSpec IFU (Obs 2)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 72.092 Arcsec (larger than the recommended limit of 60.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser. (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	HD-30365-PSF-CALIBRATOR	RA: 04 47 17.9234 (71.8246808d) Dec: +05 47 17.59 (5.78822d) Equinox: J2000			Proper Motion RA: 14.674000000000001 mas/yr Proper Motion Dec: -17.857999955595005 mas/yr Parallax: 0.005368" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[A stars, Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	4 OFFSET-TA-STAR-FOR-HD-30365	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)	Tile Order
	5	1	25.0			0.0			-20.0		0.0	ROW_ORDER
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	CYCLING			SMALL	1			9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 9535 - Observation 3 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 3: HD 15745 Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	HD-15745	RA: 02 32 55.8109 (38.2325454d) Dec: +37 20 1.04 (37.33362d) Equinox: J2000			Proper Motion RA: 46.436 mas/yr Proper Motion Dec: -47.90300001786818 mas/yr Parallax: 0.0139411" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, F stars]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	6 OFFSET-TA-STAR-FOR-HD-15745	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	2	1	-30.0		0.0		-15.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Proposal 9535 - Observation 3 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 218 to 238 Degrees (V3 79.02835083 to 99.02835083) Sequence Observations 3, 4, Non-interruptible
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Proposal 9535 - Observation 4 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 4: HD 15745 PSF Calibrator - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(7)	HD-19301-PSF-CALIBRATOR	RA: 03 07 32.0857 (46.8836904d) Dec: +39 05 12.14 (39.08671d) Equinox: J2000			Proper Motion RA: -27.145 mas/yr Proper Motion Dec: -28.295000083744526 mas/yr Parallax: 0.0100366" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	8 OFFSET-TA-STAR-FOR-HD-19301	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	2	1	-30.0			0.0		-15.0		0.0		DEFAULT
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 9535 - Observation 5 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 5: HD 15115 Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(HD 15115 Debris Disk - NIRSpec IFU (Obs 5)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 62.640 Arcsec (larger than the recommended limit of 45.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(9)	HD-15115	RA: 02 26 16.2458 (36.5676908d) Dec: +06 17 33.19 (6.29255d) Equinox: J2000			Proper Motion RA: 88.013 mas/yr Proper Motion Dec: -50.41899994466803 mas/yr Parallax: 0.020502600000000003" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	10 OFFSET-TA-STAR-FOR-HD-15115	WATA	SUB2048	F110W	NRSRAPIDD6	3	1	1	14.452	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	0.0		-30.0		0.0		20.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Proposal 9535 - Observation 5 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 204 to 224 Degrees (V3 65.02835083 to 85.02835083) Sequence Observations 5, 6, Non-interruptible
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Proposal 9535 - Observation 6 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 6: HD 15115 PSF Calibrator - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(HD 15115 PSF Calibrator - NIRSpec IFU (Obs 6)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 64.398 Arcsec (larger than the recommended limit of 45.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(11)	HD-16647-PSF-CALIBRATOR	RA: 02 40 15.6918 (40.0653825d) Dec: +06 06 43.02 (6.11195d) Equinox: J2000				Proper Motion RA: 49.263 mas/yr Proper Motion Dec: -1.2180000112493872 mas/yr Parallax: 0.0254832" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	12 OFFSET-TA-STAR-FOR-HD-16647	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	2	0.0			-30.0		0.0		20.0		DEFAULT
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 9535 - Observation 7 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 7: HD 141943 Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 7:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(13)	HD-141943	RA: 15 53 27.2917 (238.3637154d) Dec: -42 16 0.71 (-42.26686d) Equinox: J2000			Proper Motion RA: -43.236 mas/yr Proper Motion Dec: -65.58800002949283 mas/yr Parallax: 0.0166271" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, G stars]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	14 OFFSET-TA-STAR-FOR-HD-141943	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	0.0		-16.7		0.0		20.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Proposal 9535 - Observation 7 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 70 to 90 Degrees (V3 291.02835083 to 311.02835083) Sequence Observations 7, 8, Non-interruptible
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Proposal 9535 - Observation 8 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 8: HD 141943 PSF Calibrator - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(15)	HD-141885-PSF-CALIBRATOR	RA: 15 52 52.0812 (238.2170050d) Dec: -37 23 54.91 (-37.39859d) Equinox: J2000			Proper Motion RA: 3.7680000000000002 mas/yr Proper Motion Dec: -30.806999984633876 mas/yr Parallax: 0.0096624" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[G stars, Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	16 OFFSET-TA-STAR-FOR-HD-141885	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order	
	1	2	0.0			-16.7		0.0		20.0	DEFAULT	
Dithers	#	Dither Type			Size	Starting Point		Number of Points		Points		
	1	CYCLING			SMALL	1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Special Requirements	Sequence Observations 7, 8, Non-interruptible
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Proposal 9535 - Observation 15 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 15: HD 61005 Debris Disk - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(21)	HD-61005	RA: 07 35 47.4624 (113.9477600d) Dec: -32 12 14.05 (-32.20390d) Equinox: J2000			Proper Motion RA: -55.05 mas/yr Proper Motion Dec: 74.319 mas/yr Parallax: 0.0274344" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, G dwarfs]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	22 OFFSET-TA-STAR-FOR-HD-61005	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	5	0.0		35.0		0.0		30.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Proposal 9535 - Observation 15 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 166 to 186 Degrees (V3 27.02835083 to 47.02835083) Sequence Observations 15, 16, Non-interruptible
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Proposal 9535 - Observation 16 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 16: HD 61005 PSF Calibrator - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(23)	HD-56161-PSF-CALIBRATOR	RA: 07 14 36.7020 (108.6529250d) Dec: -30 39 17.63 (-30.65490d) Equinox: J2000			Proper Motion RA: -83.772 mas/yr Proper Motion Dec: 45.286 mas/yr Parallax: 0.0082534" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	24 OFFSET-TA-STAR-FOR-HD-56161	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order
	1	5	0.0			35.0		0.0		30.0		ROW_ORDER
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Special Requirements	Sequence Observations 15, 16, Non-interruptible
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Proposal 9535 - Observation 17 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 17: HD 61005 Debris Disk - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(21)	HD-61005	RA: 07 35 47.4624 (113.9477600d) Dec: -32 12 14.05 (-32.20390d) Equinox: J2000			Proper Motion RA: -55.05 mas/yr Proper Motion Dec: 74.319 mas/yr Parallax: 0.0274344" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, G dwarfs]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	22 OFFSET-TA-STAR-FOR-HD-61005	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)	Tile Order
	1	5	0.0			35.0			0.0		20.0	ROW_ORDER
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		9			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Proposal 9535 - Observation 17 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 186 to 206 Degrees (V3 47.02835083 to 67.02835083) Sequence Observations 17, 18, Non-interruptible
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Proposal 9535 - Observation 18 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 18: HD 61005 PSF Calibrator - NIRSpec IFU											Fri Mar 13 17:00:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(23)	HD-56161-PSF-CALIBRATOR	RA: 07 14 36.7020 (108.6529250d) Dec: -30 39 17.63 (-30.65490d) Equinox: J2000			Proper Motion RA: -83.772 mas/yr Proper Motion Dec: 45.286 mas/yr Parallax: 0.0082534" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	24 OFFSET-TA-STAR-FOR-HD-56161	WATA	SUB32	F110W	NRSRAPID	3	1	1	0.08	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	5	0.0		35.0		0.0		20.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Special Requirements	Sequence Observations 17, 18, Non-interruptible
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Proposal 9535 - Observation 11 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 11: AU Mic Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(AU Mic Debris Disk - NIRSpec IFU (Obs 11)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 52.877 Arcsec (larger than the recommended limit of 50.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 11:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(25)	AU-MIC	RA: 20 45 9.5325 (311.2897188d) Dec: -31 20 27.24 (-31.34090d) Equinox: J2000			Proper Motion RA: 281.319 mas/yr Proper Motion Dec: -360.1480000497759 mas/yr Parallax: 0.1029432" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, Exoplanet Systems, M dwarfs]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	26 OFFSET-TA-STAR-FOR-AU-MIC	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	5	0.0		20.0		0.0		15.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Proposal 9535 - Observation 11 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 36 to 62 Degrees (V3 257.02835083 to 283.02835083) Sequence Observations 11, 12, Non-interruptible
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Proposal 9535 - Observation 12 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 12: AU Mic PSF Calibrator - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(AU Mic PSF Calibrator - NIRSpec IFU (Obs 12)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 57.154 Arcsec (larger than the recommended limit of 55.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(27)	HD-192855-PSF-CALIBRATOR	RA: 20 18 6.0466 (304.5251942d) Dec: -26 13 56.33 (-26.23231d) Equinox: J2000			Proper Motion RA: 8.631 mas/yr Proper Motion Dec: 3.01 mas/yr Parallax: 0.001062" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Calibration Description=[Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	28 OFFSET-TA-STAR-FOR-HD-192855	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	5	0.0		20.0		0.0		15.0		ROW_ORDER	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	1	false	true	NONE	9	9	787.8	

Special Requirements	Sequence Observations 11, 12, Non-interruptible
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Proposal 9535 - Observation 13 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 13: HD 202917 Debris Disk - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 13:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(17)	HD-202917	RA: 21 20 49.9576 (320.2081567d) Dec: -53 02 3.16 (-53.03421d) Equinox: J2000			Proper Motion RA: 31.426999999999996 mas/yr Proper Motion Dec: -95.58700010074972 mas/yr Parallax: 0.021409300000000003" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, G dwarfs]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	18 OFFSET-TA-STAR-FOR-HD-202917	WATA	SUB32	F110W	NRSRAPIDD6	3	1	1	0.26	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	0.0		-16.7		0.0		20.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Proposal 9535 - Observation 13 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Special Requirements	Aperture PA Range 33 to 53 Degrees (V3 254.02835083 to 274.02835083) Sequence Observations 13, 14, Non-interruptible
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Proposal 9535 - Observation 14 - The Next Step On the Water Trail: Understanding the Varied Lifecycles of Volatile Ices in Debris Disks

Observation	Proposal 9535, Observation 14: HD 202917 PSF Calibrator - NIRSpec IFU										Fri Mar 13 17:00:52 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(HD 202917 PSF Calibrator - NIRSpec IFU (Obs 14)) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 50.293 Arcsec (larger than the recommended limit of 50.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser. (Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(19)	HD-199201-PSF-CALIBRATOR	RA: 20 58 11.9458 (314.5497742d) Dec: -58 30 18.58 (-58.50516d) Equinox: J2000			Proper Motion RA: -40.535 mas/yr Proper Motion Dec: -81.06800003133685 mas/yr Parallax: 0.0117922" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Calibration Description=[G stars, Point spread function]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	20 OFFSET-TA-STAR-FOR-HD-199201	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	225266	
Template	HFF Readout Mode											
	false											
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	0.0		-16.7		0.0		20.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	

Special Requirements	Sequence Observations 13, 14, Non-interruptible
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